

Institute of Actuaries of India

May 2010 EXAMINATION

Subject ST3 — General Insurance

Specialist Technical

Indicative Solution

1).

i) The two main types of proportional reinsurance are quota share and surplus reinsurance.

ii) **Quota share:** assume 35% is retained. If the claim is for Rs 60,000 the direct writer pays Rs 21,000 and the reinsurer pays Rs 39,000.

Surplus: assume the retention limit is Rs 100,000 and the reinsurer's maximum cover is Rs 400,000. If the sum insured is Rs 120,000, the direct writer's share is Rs 100,000 and the reinsurer's share is Rs 20,000.

If the claim is Rs 60,000, the direct writer retains Rs 50,000 and the reinsurer pays Rs 10,000.

If the sum insured is Rs 550,000 the direct writer's share is Rs 150,000 and the reinsurer's share is Rs 400,000.

If the claim is Rs 60,000, the direct writer retains Rs 16,364 and the reinsurer pays Rs 43,636

[4]

2).

i) The adjustment coefficient equation is $\lambda M_x(r) = \lambda + cr$

For the direct insurer, the net value c is

$$c = (1 + \theta_1) \lambda m - (1 + \theta_2)(1 - k) \lambda m$$

$$c = [k(1 + \theta_2) - (\theta_2 - \theta_1)] \lambda m$$

The moment generating function of the net claim amount X is

$$M_x(r) = E[\exp(rX)] = e^{rkm}$$

So the adjustment coefficient (after dividing through by λ) becomes:

$$1 + [k(1 + \theta_2) - (\theta_2 - \theta_1)]mr = e^{rkm}$$

ii) Differentiate w.r.t. 'k' given

$$(1 + \theta_2)mr + [k(1 + \theta_2) - (\theta_2 - \theta_1)]m \frac{dr}{dk} = (rm + km \frac{dr}{dk})e^{rkm}$$

Divide through by m

$$(1 + \theta_2)r + [k(1 + \theta_2) - (\theta_2 - \theta_1)] \frac{dr}{dk} = (r + k \frac{dr}{dk})e^{rkm}$$

At the max, $\frac{dr}{dk} = 0$

dk

$$\text{So, } (1 + \theta_2)r = re^{rkm}$$

If $r > 0$, we cancel the r 's

$$(1 + \theta_2) = e^{rkm} \text{ i.e. } mr = \log(1 + \theta_2)/k$$

Substituting for mr on the LHS and e^{rkm} on the RHS of equation (1) in part (i)

$$1 + [k(1 + \theta_2) - (\theta_2 - \theta_1)] \log(1 + \theta_2)/k = (1 + \theta_2)$$

Canceling the 1's and re-arranging,

$$k = (\theta_2 - \theta_1) / [1 + \theta_2 - \theta_2 / \log(1 + \theta_2)]$$

iii) With $\theta_1 = 0.2$, $\theta_2 = 0.35$ and $m = 5000$ the optimal value of k is 0.8164

The corresponding value of r can be found from $mr = \log(1 + \theta_2)/k$

$$r = 0.00007352$$

Using Lundberg's inequality, the approximate probability of ruin is less than or equal to $e^{-rU} = e^{-5.51} = 0.004$

[8]

3).

The main aim of the company is to maximize investment returns subject to meeting liabilities as they fall due. Hence the company would want to select investments which match the liabilities as closely as possible.

The level of free reserves determines the extent to which the company can depart from a matched strategy.

Since the company is rapidly growing, the size of free reserves relative to written premium and statutory solvency may be under pressure.

Different matching considerations for vehicle damage and bodily injury claims.

Vehicle damage claims are short tailed i.e claims are reported and settled quickly. Hence cash, with its liquidity and stable capital value, would be a suitable investment to match vehicle damage claims.

Bodily injury claims are long tailed and hence assets which provide protection against inflation are necessary. Equities and index linked securities provide some inflation protection but inflation affecting bodily injury claims in particular will not necessarily correspond with the type of inflation underlying the index (in the case of index linked government securities). Cash may not be a suitable investment for bodily injury claims as it does not provide protection against inflation.

Equities are expected to produce higher returns than index linked securities but have greater volatility. This is not suitable if solvency is under pressure. Risk of default with equities is higher as compared with cash and index linked government securities.

Regulatory restrictions will have an impact on the types and volumes of assets which can be held. The taxation rules for different assets may affect investment decisions.

While cash, equities and index linked securities are marketable, the level of security and appropriateness depends on the mix of these assets and the currency of the assets and liabilities.

[8]

4).

$$i) \quad OP = RP + cf*CE + v*OP + j + I + E$$

RP is the risk premium

cf*cE is the claims related expenses such as legal fees

v*OP is the variable premium related expenses i.e. commission

j is the variable per policy expenses such as variable administration costs

E is the contribution to profit and fixed costs e.g. staff salaries, rents

I is the additional initial costs associated with writing a new policy which is spread over the lifetime of a policy

The average expected rate of commission should be used.

When loading for expenses the aim is to determine the premium for each class sufficient to cover the expenses attributable to that class.

Fixed costs are recouped as an amount per policy or a proportion of the premium or a combination of both. Similarly for profit. The expected

volumes of business can be used to determine the amount of loadings for fixed costs

Other loadings: The risk premium should be loaded to allow for the cost of reinsurance (depending on the reinsurance arrangements in place).

There may be a contingency loading added to the risk premium to reflect the degree of uncertainty in the risk premium. This loading will be implicitly covered by a higher profit loading.

Investment income can be allowed for by an adjustment to the bottom

line profit figure (with a lower contribution to profit) or may also be allowed for explicitly e.g. by subtraction of investment income or using discounted assets in the formula.

- ii) The office premium is calculated as $1.3P$ where P denotes the risk premium which is calculated using the following credibility formula:

—

$$P = ZX + (1-Z)\mu$$

Using the values given, the estimated risk premiums are

$$P_p = 0.5 * 0.8\mu + 0.5\mu = 0.9\mu$$

$$P_q = 0.75 * 1.2\mu + 0.25\mu = 1.15\mu$$

So the ratio of risk premiums will be $P_p / P_q = 0.9 / 1.15 = 0.7826$

Office premiums for risk P is 78.26% of the office premium for risk Q.

[9]

5). Solution

- i) In one year accounting method, a proportion of the premiums received in a given financial year is reserved to meet the liability arising from the exposure after the accounting year end. This reserve is called Unearned Premium Reserve (UPR).

If a calculation of the reserve is done on a prospective basis based on past claim experience and expense levels (rather than simply taking premium) to estimate the required amount to cover the unexpired exposure, this prospective amount is called Unexpired Risk Reserve (URR).

If $URR > UPR$, then UPR will be inadequate to cover the unexpired risk and expenses.

In such circumstances an additional reserve the AURR, or additional unexpired risk reserve will be required if full provision is to be made for future liabilities.

- ii) Assumptions

- a. Policies are annual and risk is even over each policy year
- b. Policies commence mid-period on average.
- c. Expense ratio remains same after the premium rate increase.
- d. Expense ratio applies on earned premium
- e. Mix of business remains same after the premium increase
- f. The calculation of UPR and URR is on annual basis to calculate the AURR for the year (i.e. credit is taken for the quarter where $URR < UPR$)

Calculations

Combined Ratio at the new rate = $114/120 = 95\%$.

We can use Eighths method to approximate the UPR for quarter 1,2 and 4. For Quarter 3 a separate calculation will be needed.

Period	Written Premium	Unearned Exposure	UPR	Combined Ratio	URR
Q1	100	1/8	12.50	114%	14.25
Q2	120	3/8	45.00	114%	51.30
Q3@Old Rate	100	13/24	54.17	114%	61.75
Q3@New Rate	50	8/12	33.33	95%	31.67
Q4	120	7/8	105.00	95%	99.75
Total			250.00		258.72

So AURR = Rs 8.72 Crore

- iii) Three reasons why change in combined ratio may be significantly different from a decrease of 16.67% as assumed above:
- The mix of business may change wherein bad risks will stay in the portfolio and good risks may leave. The reduction in combined ratio will be lower in such a case.
 - The competitors may have increased their rates as well. The impact on the mix of business and hence the impact on combined ratio will depend on the change in premium rates by the competitors. Expense ratio (as per definition) is calculated based on written premium. The increase in premium rates may lower the premium, but management expenses may not decrease in the same proportion. So the reduction in combined ratio will be lower.
 - To keep the business volume at the earlier level, the acquisition expenses may need to be increased. This will also increase the combined ratio.
 - Other external factors such as
 - Unexpected large losses (or catastrophic events)
 - Actual inflation significantly different from expected claim inflation

[13]

6).

- i) The main areas of risk and uncertainty surrounding claims experience include:
- Variability in the amount and number of claims
 - Differences in the choice of valuation bases
 - Differences in target markets which influence the attitude of policyholders towards claiming including anti-selection
 - Differences in policy wordings and exclusions to cover which will have an impact on the nature and quantum of claims
 - Changes in legislation, judicial decisions and court award inflation which affects liability claims
 - Differences in reinsurance arrangements which affect the net retention and likelihood of reinsurer default
 - Claims handling efficiency which may lead to reporting or settlement delays

The main areas of risk and uncertainty surrounding expense assumptions:

Different mix of business from that expected which may impact projected business plan

- Expense loadings may be different given the high set up costs and these will need to be amortised suitably
- Uncertainty whether allocation of expenses by type of business is appropriate
- Higher than expected legal and professional charges
- Higher than expected inflation of the kind that affects expenses like salary inflation
- Lower than expected business volumes resulting in higher per policy expenses
- If different commission rates are charged on business acquisition and renewal, there is a persistency risk relating to the spread of differential rates of commission over future levels of business

The main areas of risk and uncertainty surrounding investment income:

- Market conditions may worsen against the securities in which the insurer is invested in
- Portfolio likely to be different from the industry i.e. no claims outgo at the outset
- Investible funds may be lower than expected if claims are paid sooner than expected
- A sizeable portion of the assets may not be available for investment i.e. delay in premiums

- ii) The issues that this newly formed company should consider when assessing the security of a reinsurer are:

Being a new entrant to the market, this insurer will look to the reinsurer for technical expertise when deciding on premium rates, policy terms and conditions. The insurer should assess the technical sophistication of the reinsurer through its market reputation and market views on security and pricing adequacy.

The mix of business being written i.e. long tail or short tail business, as greater security is required for long tail classes of business, given the lay in emergence of claims.

The reinsurer's solvency position to ensure that the reinsurer will still be around when claims are to be paid.

Domicile of the reinsurer and the supervisory/regulatory regime where domiciled.

The reinsurer's financial statements to ensure that its assets are adequate to meet its liabilities and its technical reserves are adequate Ownership of reinsurer

Quality of management

Global presence and size of the reinsurer to ensure that its liabilities are spread by class and geographically to minimize exposure to catastrophes Credit rating of the reinsurer as there may be regulatory requirements to use reinsurers with a specified rating.

[13]

7).

- i) The homeowner may under declare the value of contents at proposal or may fail to revise the figure through time where affected by inflation.

The insured may fail to notify the insurer of changes in personal circumstances relevant for assessing the risk e.g. house may be empty during the day.

The insured may have provided misleading information regarding security measures at the premises.

The insured may submit false claims or deliberately inflate valid claims by including items that were never part of the cover or never damaged

The insured may behave carelessly as a result of taking out insurance i.e. leaving the house without locking doors/windows.

The insured may withhold risk related information on the proposal form i.e. past claims, fraud, arson and deliberate damage

The insured may under declare the rebuilding cost of the house at proposal. This may simply be through failing to provide accurate information at purchase or may be due to failing to advise the insurer

where modifications have taken place i.e. renovation work, extensions etc.

ii) For the applicant

Advantages:

There is no need to estimate the value at risk and to keep it up to date with inflation. This would be difficult without the assistance of a professional estimator/surveyor

The applicant is sure that he has purchased adequate insurance cover.

Individuals with overly valuable goods in a small house would be more likely to purchase cover as the premium will be lower than that offered by other companies.

Disadvantages:

Individuals living in large houses with many rooms unoccupied (ancestral property) may be over-insured and will pay higher premium than offered by other companies.

Given the ambiguity in the product, individuals living in identically structured houses may be required to pay different premiums if one had undertaken renovation work for example and removed an adjoining wall which resulted in fewer rooms in the house.

For the company

Advantages:

There will be no need to assess the sum at risk to ensure that cover is adequate. This will reduce claims management expenses.

There is reduced risk of adverse publicity if as a result of under insurance the company scales down the claim amount.

Prospective policyholders may find it simpler to fill in the proposal form as they would not need to assess the value of contents. This could be a potential marketing advantage.

Disadvantages:

This being a new product, the company will not have its own data on which to price the portfolio. There is a risk of getting the premiums wrong. This is compounded by the fact that the company is moving away from the market's rating approach.

There is increased risk of anti-selection as individuals with high value contents living in small houses will be more likely to insure with this company owing to lower premiums. Conversely for individuals with low value contents in large households. As a result of this anti-selection, claims experience may be worse than expected.

There is a risk that the applicant may not correctly state the number of rooms in the property i.e. exclude bathrooms, store rooms or may not provide details of renovation work undertaken which would have altered the room layout. This would lead to inadequate premiums being charged.

iii) To mitigate the company's risks, it could:

- Reduce ambiguity in the proposal form by clearly specifying the definition of 'room' i.e. whether it includes bathrooms etc
- Specify the maximum limit of single articles and limit the value of jewelry and other valuables
- Regularly monitor rates against those of competitors to ensure adequacy of premiums being charged
- Increase the excess limits.

[13]

8). i). The rating factors primarily used for pricing motor insurance include:

- make and model of the vehicle
- age of the driver
- vehicle use
- additional drivers
- stated mileage
- excess
- gender of main driver
- occupation of driver
- claims record/NCD
- driving restricted to named drivers
- security features
- modifications
- covered parking

ii) Two risk factors which are used to price personal motor insurance but which cannot be used as rating factors are:

- Ability of driver
- Density of traffic in which vehicle is driven

These risk factors are not objectively measurable and easily quantifiable and hence they cannot be used as rating factors.

- iii) The mix of claims may have changed in recent years i.e. more bodily injury claims than own damage. A larger proportion of the bodily injury claims may be settled in court which would involve additional costs such as legal fees.

There may have been staffing changes in the claims department as a result of which more high cost resources may be allocated to servicing claims. Alternatively claims volumes may have fallen but claims department staffing remained unchanged.

The mix of business may have changed i.e. the company may be writing more high valued vehicles which give rise to larger than average claims. Claims expenses as a percentage of claims may be greater for larger claims.

The company may have increased its budget for claims management to ensure that fraudulent or overstated claims were minimized /eliminated.

Thus while claims management expenses as a percentage of claim costs have increased the overall loss ratio may have reduced.

The company may have not reviewed its thresholds (in line with inflation) for surveyors'/loss adjustors' triggers. As a result greater number of claims would call for a loss adjustor thereby leading to higher expenses.

- iv) Claims department staffing levels and/or the efficiency of staff could be reviewed.

Operational improvements for efficiency could be introduced

The company could review any fixed limits for claims management in line with inflation. This however may increase the number of overstated claims and thus increase rather than reduce overall costs.

- v). The movements which are normally monitored are:

- New business
- Renewals/Lapses
- Endorsements
- Mid-term cancellations

By studying portfolio movements and their trends, an insurer can:

- Measure the growth or contraction of the business and of its different segments
- Get an early indication of unusual losses or gains in business volumes. These may indicate if the insurer's risks or conditions are out of line with the market
- Assess the effects of a new set of rates or a marketing campaign on the business and thus the sensitivity of the portfolio to market forces.
- Movements will give an early indication of adverse or favorable changes. These may indicate the need for a review of premiums or terms and conditions. These may also be used to help reconcile accounts and in-force statistics

9).

- i) In the simple proportion method, IBNR is estimated simply as a proportion of a figure that appears in the accounts such as earned premium, paid claims or outstanding claims for a line of business.

The figures from the accounts can be taken from the annual accounts or more frequent internal accounts.

The ratio can be derived using the past experience or from an industry standard.

This approach can be used:

- a. where the class is not significant in size when compared with all other classes
- b. where the class is very short tailed
- c. as an approximate check on another (more complex) method.

In Delay Pattern method, the number of IBNR claims and the average IBNR claim amounts are estimated separately and IBNR reserve is calculated as

$$\Sigma \text{ Expected number of IBNR claims} \times \text{ Expected average cost per IBNR claim}$$

The expected number of IBNR claims can be estimated using the cumulative number of claim run-off triangle by applying statistical methods such as the chain-ladder method.

The average amount per claim can be estimated using the reported claims. The past relationship between average claim amount of reported claims and average claim amount of IBNR claims can be applied.

The estimation of average amount of IBNR claims should allow for future inflation and other trends, known features or changes in business.

This method can be used:

- a. where there is enough experience to estimate/build a reliable number of claim run-off triangle
- b. for lines of business where claim reporting is not very long tail and the number of claims reported follow a pattern

- ii)

Cumulative triangle

	Lag in quarter					
Occurrence Quarter	1	2	3	4	5	6
Q1 2008	12	17	17	19	19	21
Q2 2008	15	22	24	27	28	
Q3 2008	12	13	14	18		
Q4 2008	59	66	69			
Q1 2009	46	52				
Q2 2009	55					

Sum of rows	199	170	124	64	47	21
Sum excluding last	144	118	55	46	19	
Development factor	1.181	1.051	1.164	1.022	1.105	

Projected claims						
Q1 2009		54.644	63.586	64.968	71.807	
Q2 2009		64.931	68.232	79.397	81.123	89.663

Number of IBNR claims by Lag						
Q1 2009		2.644	8.942	1.382	6.839	
Q2 2009		9.931	3.302	11.165	1.726	8.539

Average Claim Size						
		102,411	104,881	107,410	110,000	112,653

IBNR Amounts						
Q1 2009		270,783	937,819	148,473	752,262	2,109,337
Q2 2009		1,017,002	346,270	1,199,259	189,863	961,974
Total (in Rs 000s)						5,824

- The first quarter is still developing and is not fully run-off. In the above calculation, no tail factor has been used which may not be appropriate.
- The number of claims is not large enough to estimate the link-ratios reliably for lag 3 or more.
- If market benchmark link ratios are available, the link ratios calculated from the data should be compared with the benchmark link ratios. The final link ratio to be used should be selected as a blend of the benchmark link ratio and the ratios calculated from the data.
- A suitable tail factor should be selected using the market benchmarks if available.
- Even if benchmarks are not available, a factor more than one should be selected based on judgment or after discussions with the claims managers.
- The link ratio from lag 1 to lag 2 for quarters 1 and 2 of 2008 are much higher than the same ratios for the subsequent quarters. The link ratio from lag 1 to lag 2 may be calculated as 3-quarter weighted average.

[16]

[Total Marks 100]
